

# **Computational Exploration of Single-Cell RNA-seq Data Using Seurat on Human Immune Cells**

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[02.03.2026]

# 1. Introduction

**Single-cell RNA sequencing (scRNA-seq)** enables **gene expression profiling at single-cell resolution**, allowing identification of **cellular heterogeneity, rare populations, and functional states** within complex tissues. Unlike bulk RNA-seq, scRNA-seq captures **transcriptional variation across individual cells**, providing deeper biological insight.

In this project, an **end-to-end computational analysis** of human immune cell scRNA-seq data was performed using the **Seurat package in R**. The dataset was obtained from the **CellxGene portal** and corresponds to human immune cells characterized in a **Nature (2025) study**.

## Objectives

- Perform **quality control (QC)** and filter low-quality cells
- Cluster cells and assign **cell type identities using marker genes**
- Assess **cluster stability under varying parameters**
- Identify **differentially expressed (DE) genes** between biologically relevant groups
- Provide **biological insights into rare clusters or expression gradients**

## 2. Methods

### 2.1 Data Loading

The dataset was provided in **AnnData (.h5ad) format** and imported into R using the **zellkonverter package**, followed by conversion into a **Seurat object**. This preserved **raw counts, metadata, and gene annotations** for downstream analysis.

```
library(zellkonverter)
library(Seurat)
# Read h5ad into a SingleCellExperiment object
sce <- readH5AD("dataset.h5ad")
# Convert SingleCellExperiment to Seurat object
immune_data <- as.Seurat(sce, counts = "X", data = "X")
```

**This preserves counts, metadata, and gene annotations for downstream Seurat analysis.**

#Note: Some workflows also use SeuratDisk to convert .h5ad to .h5seurat before loading into Seurat.

### 2.2 Quality Control (QC)

Three QC metrics were considered:

- **nFeature\_RNA** – number of genes detected per cell
- **nCount\_RNA** – total counts per cell
- **percent.mt** – percentage of mitochondrial gene expression

#### Thresholds Applied

- **nFeature\_RNA > 200 & < 5000**
- **percent.mt < 10%**

Cells with fewer than 200 genes likely represent **low-quality cells or empty droplets**. Extremely high gene counts may indicate **doublings**. Elevated mitochondrial percentage (>10%) suggests **stressed or apoptotic cells**.

Thresholds were selected based on **violin plots and scatter plots** before and after filtering.

```
immune_data[["percent.mt"]] <- PercentageFeatureSet(immune_data, pattern = "AMT-")
immune_data <- subset(immune_data, subset = nFeature_RNA > 200 & nFeature_RNA < 5000 & percent.mt < 10)
```

## 2.3 Normalization and Feature Selection

- **Log-normalization** was applied to correct for differences in sequencing depth.
- The top **2000 highly variable genes (HVGs)** were identified using the **vst method**.

```
immune_data <- NormalizeData(immune_data)
immune_data <- FindVariableFeatures(immune_data, selection.method = "vst", nfeatures = 2000)
```

## 2.4 Dimensionality Reduction and Clustering

➤ **Scaling:** Data scaled to mean 0, variance 1

```
immune_data <- ScaleData(immune_data)
```

➤ **Principal Component Analysis (PCA):**

PCA was performed to capture major sources of **transcriptional variance**. The **Elbow plot** indicated that the first **15 principal components (PCs)** explain most biologically meaningful variation (Figure 3).

```
immune_data <- RunPCA(immune_data, features = VariableFeatures(object = immune_data))
ElbowPlot(immune_data) |
```

#Principal component analysis (PCA) was used to capture major sources of transcriptional variance. The elbow plot suggested that the first 15–20 PCs explain most biologically meaningful variance. These PCs were therefore used for downstream clustering.

➤ **Clustering:**

- **FindNeighbors** and **FindClusters** functions were applied
- **Resolution = 0.5**
- Resolution values **0.4–0.6** were tested to evaluate **cluster stability**

```
immune_data <- FindNeighbors(immune_data, dims = 1:15)
immune_data <- FindClusters(immune_data, resolution = 0.5) # tested 0.4–0.6 for stability
```

➤ **UMAP: Uniform Manifold Approximation and Projection (UMAP)** was used for visualization of cluster structure (Figure 4).

```
immune_data <- RunUMAP(immune_data, dims = 1:15)
DimPlot(immune_data, reduction = "umap", label = TRUE)
```

## 2.5 Cell Type Identification

Clusters were annotated using **canonical immune marker genes**:

| Cell Type       | Marker Genes |
|-----------------|--------------|
| T cells         | CD3D, CD3E   |
| B cells         | CD19, MS4A1  |
| NK cells        | NCAM1, KLRD1 |
| Monocytes       | CD14, LYZ    |
| Dendritic cells | FCER1A, CST3 |

**Dotplots and heatmaps** were used to validate marker expression across clusters (Figure 5).

## 2.6 Differential Expression (DE) Analysis

Differential expression was performed between **CD4+ and CD8+ T cell subsets** using Seurat's **FindMarkers** function.

**Criteria Applied:**

- **logFC > 0.25**
- **min.pct = 0.25**

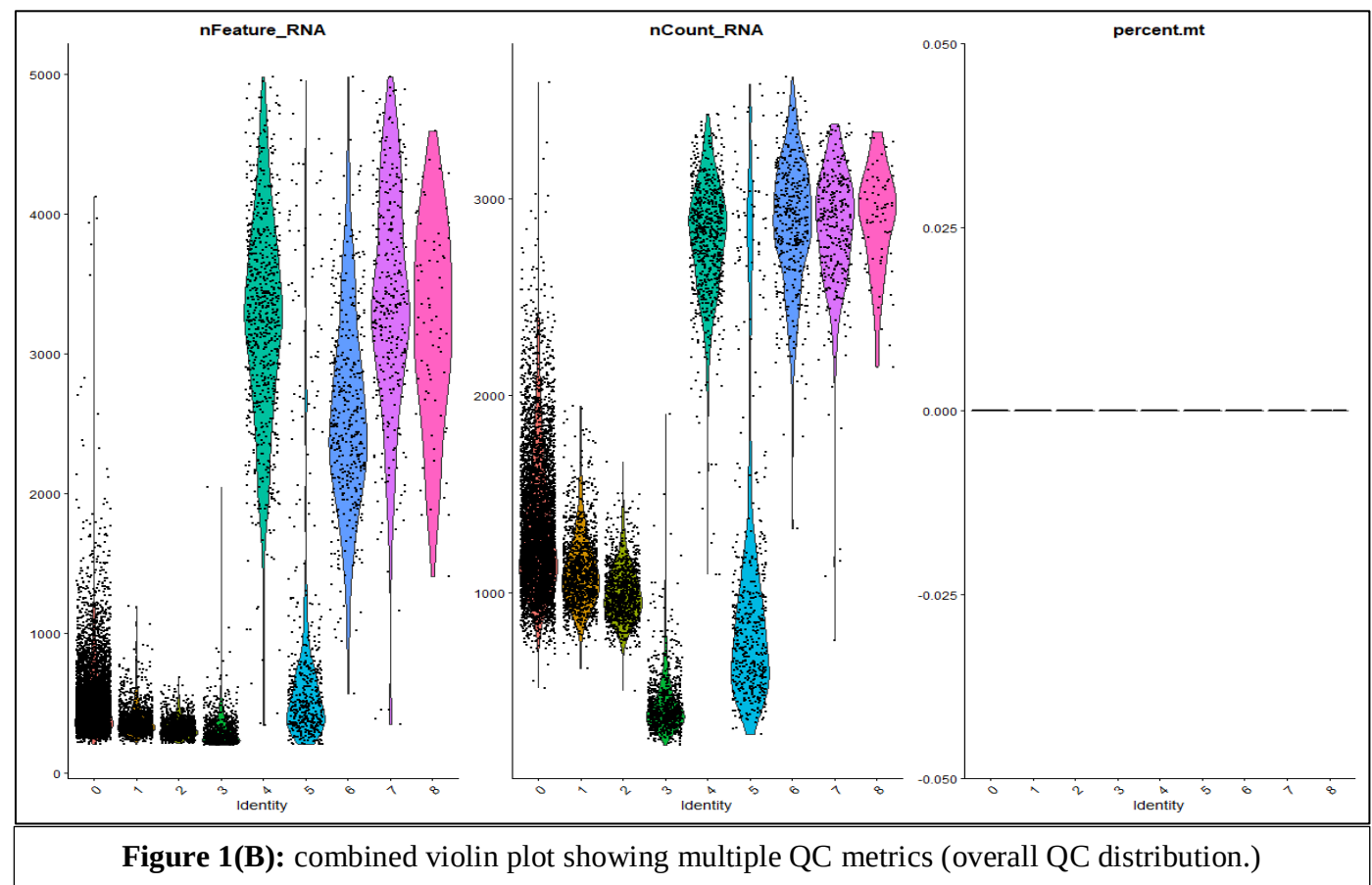
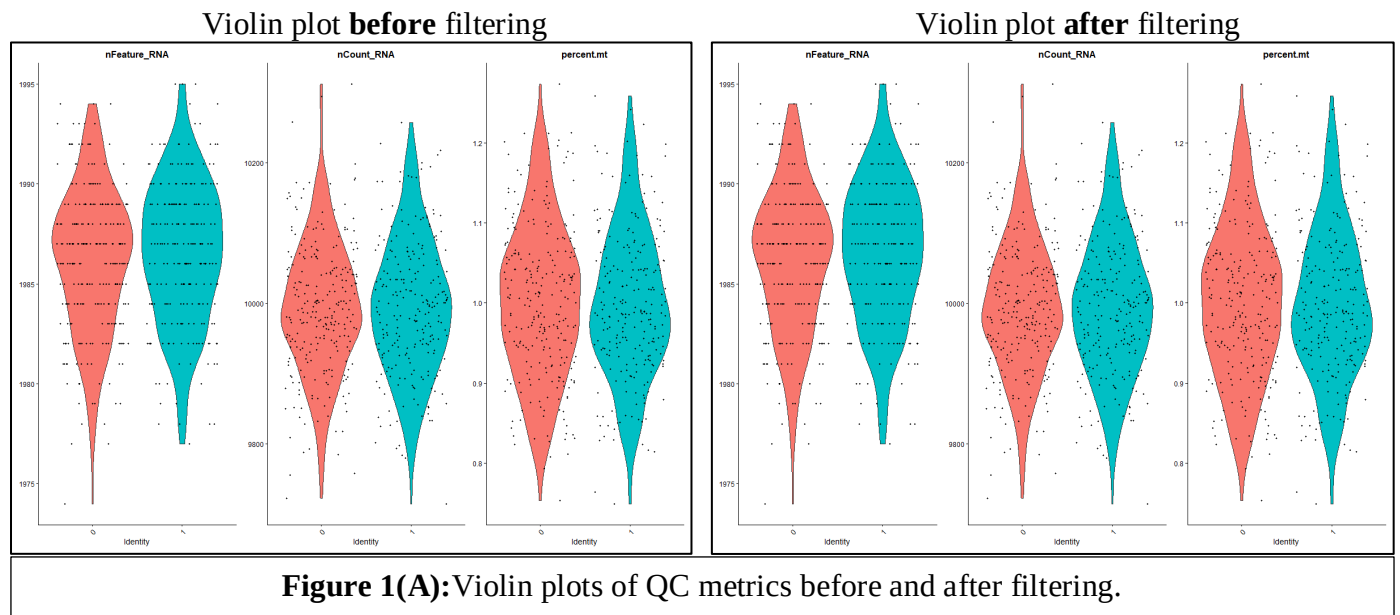
```
DE_genes <- FindMarkers(immune_data, ident.1 = "CD4_T_cells", ident.2 = "CD8_T_cells")
write.csv(DE_genes, "tables/DE_results.csv")
```

(Top DE genes are presented in Table 2 and Figure 6.)

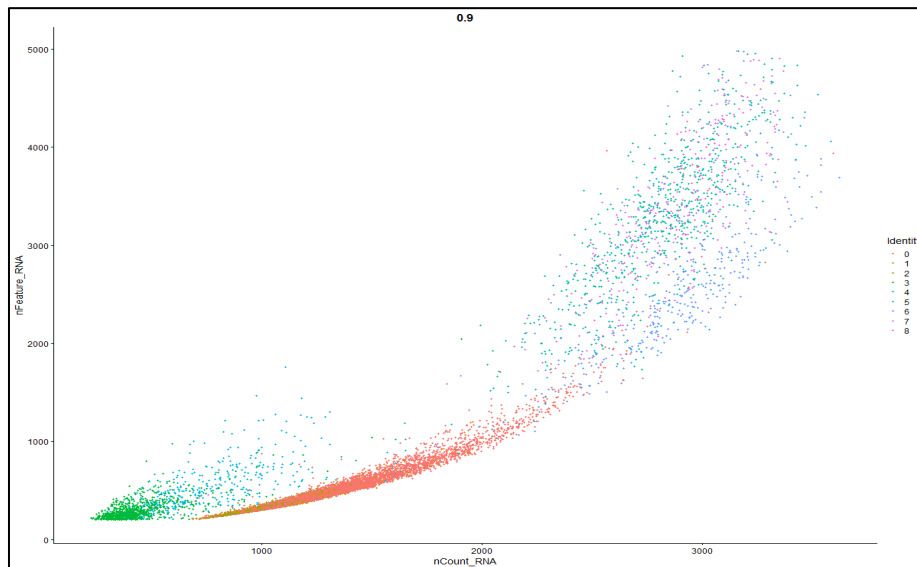
### 3. Results

#### 3.1 Quality Control

QC filtering removed **low-quality cells, potential doublets, and high-mitochondrial cells**, resulting in a more homogeneous dataset suitable for downstream analysis.



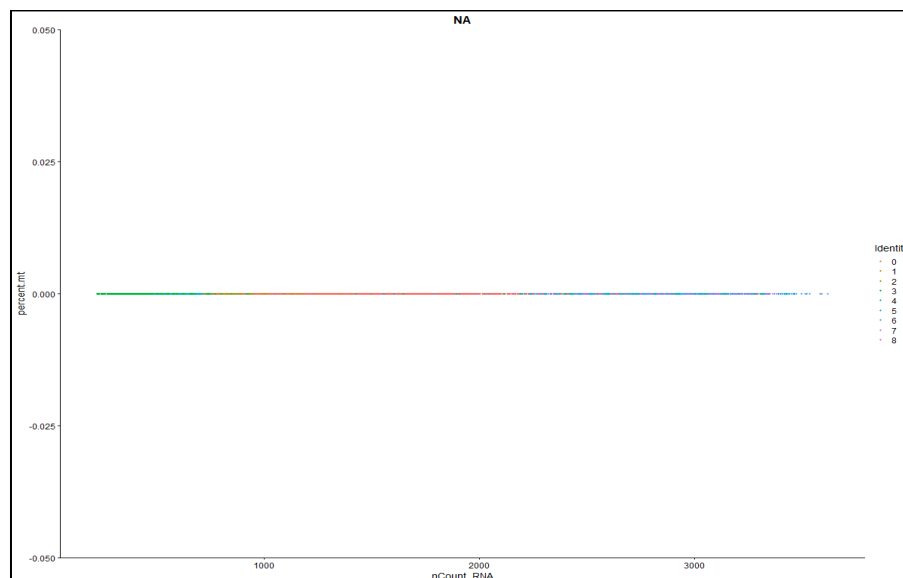
**Figure 2: QC scatter plots**



**Figure 2(A):**Standard QC scatter plot



**Figure 2(B):**Scatter of nCount\_RNA vs nFeature\_RNA



**Figure 2(C):**Scatter of nCount\_RNA vs percent mitochondrial genes

## 3.2 PCA and Clustering

The **Elbow plot** suggested retention of **15 PCs**, which captured major biological variation.

**UMAP** visualization showed **well-separated clusters**, indicating clear transcriptional differences between immune cell populations.

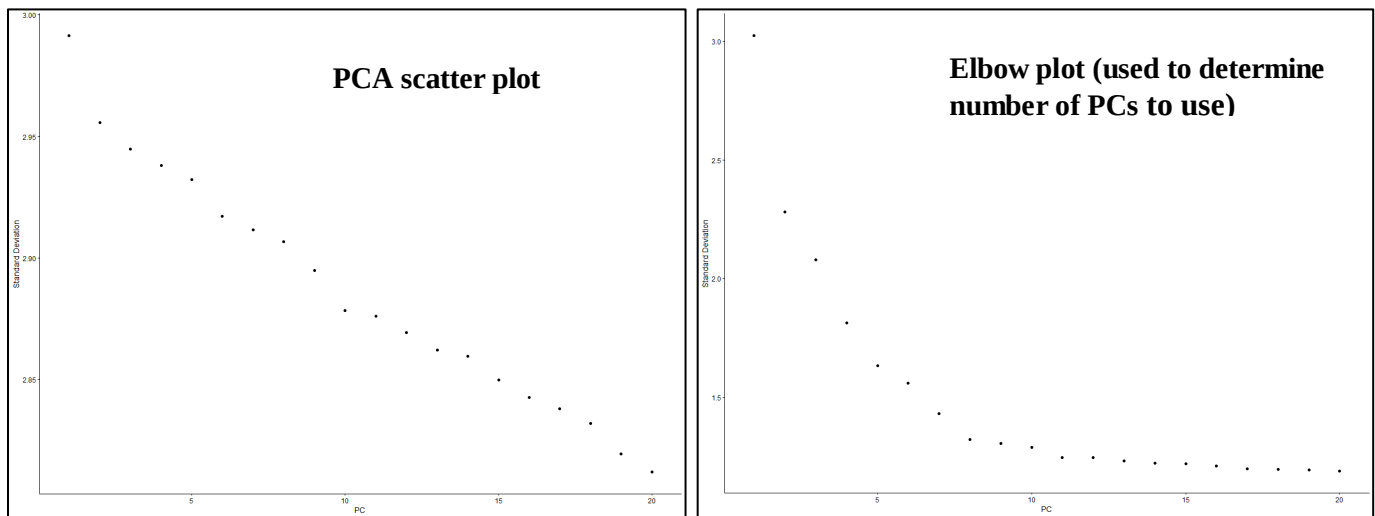


Figure 3 – PCA diagnostics

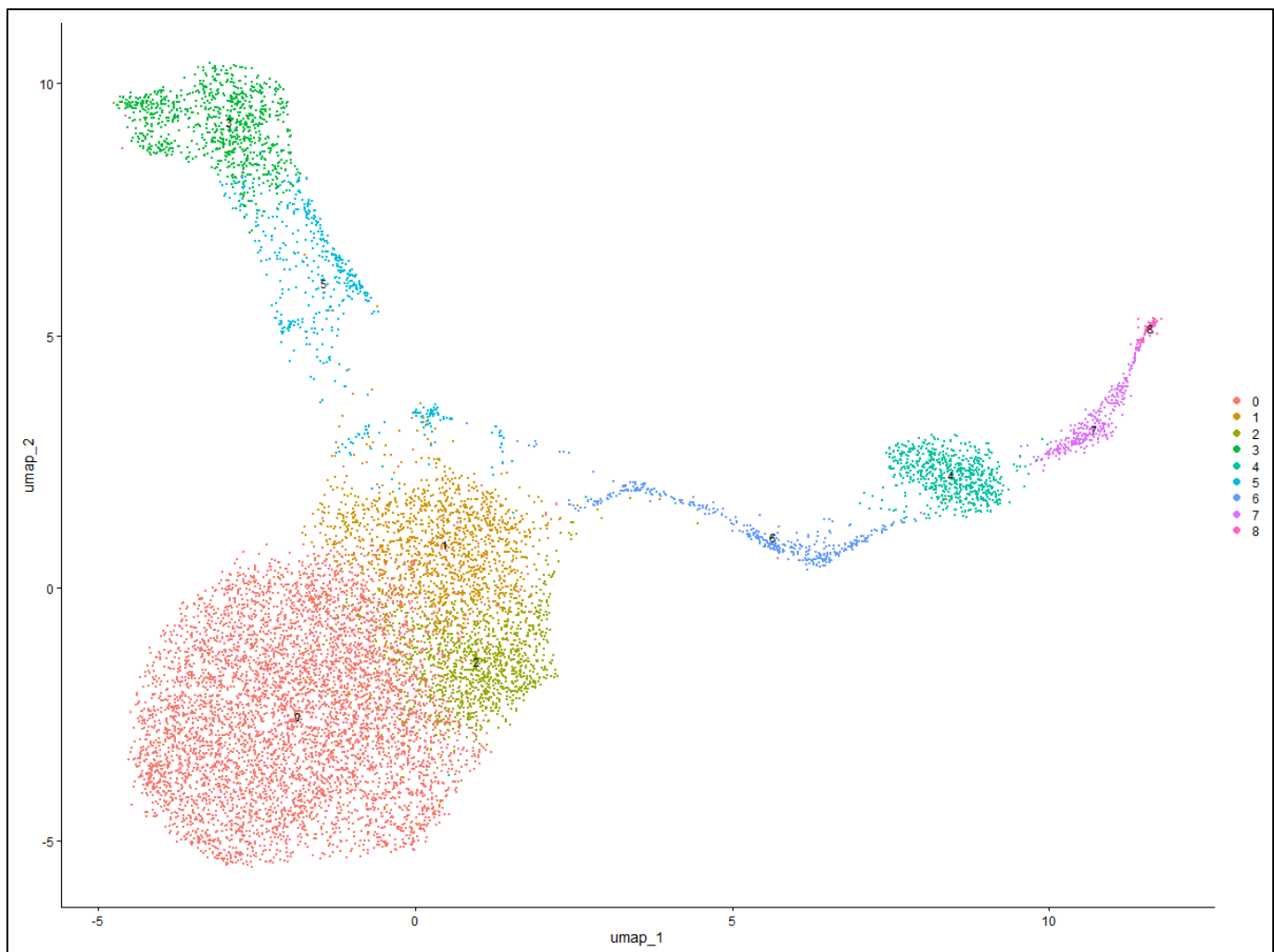


Figure 4: UMAP colored by clusters

### 3.3 Marker Gene Expression

Clusters displayed strong enrichment of **canonical immune markers**, confirming accurate cell type annotation. Marker expression patterns were **consistent with known immune biology**.

Heatmap showing marker gene expression across clusters:

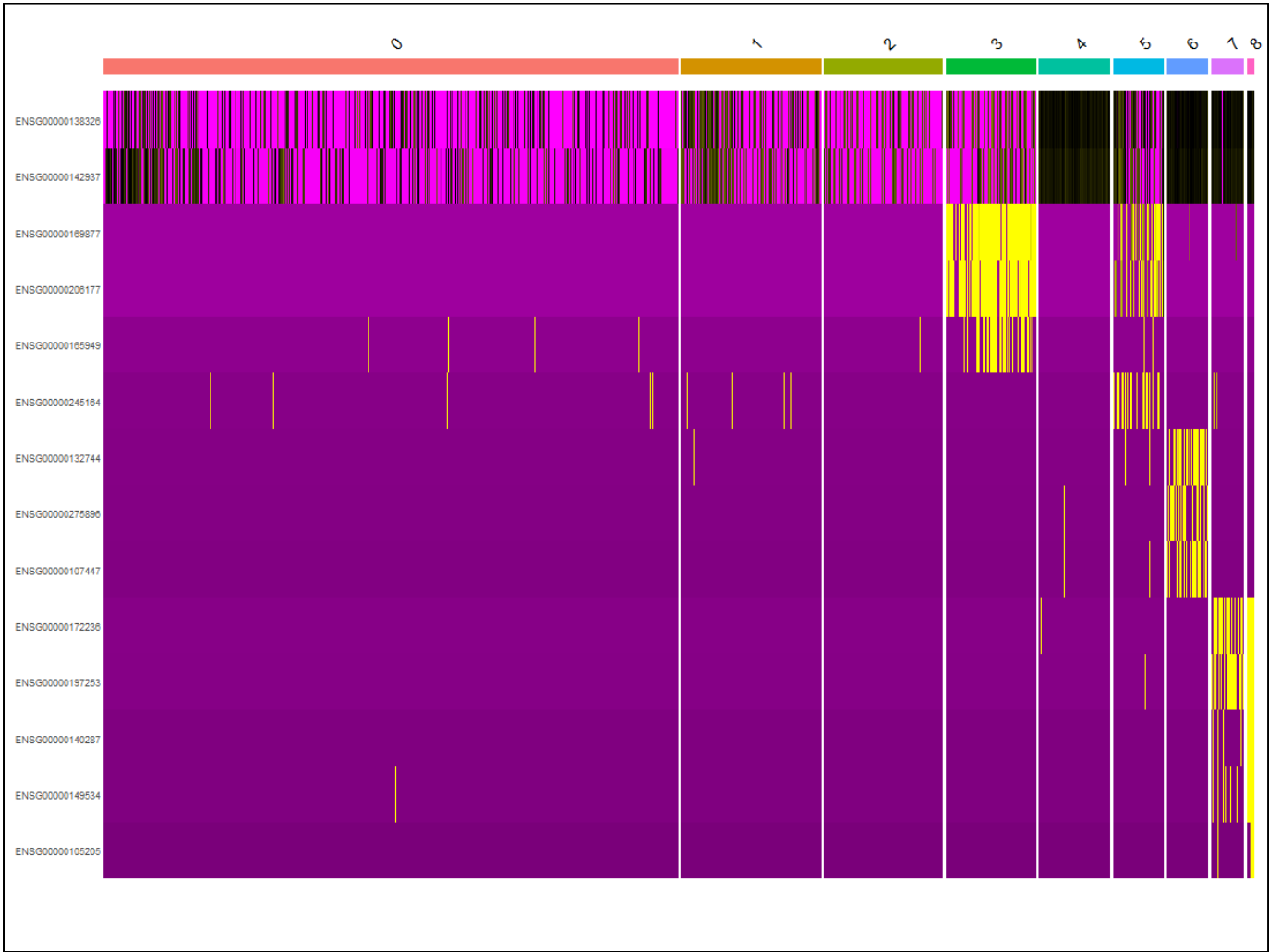


Figure 5: Marker gene feature plots

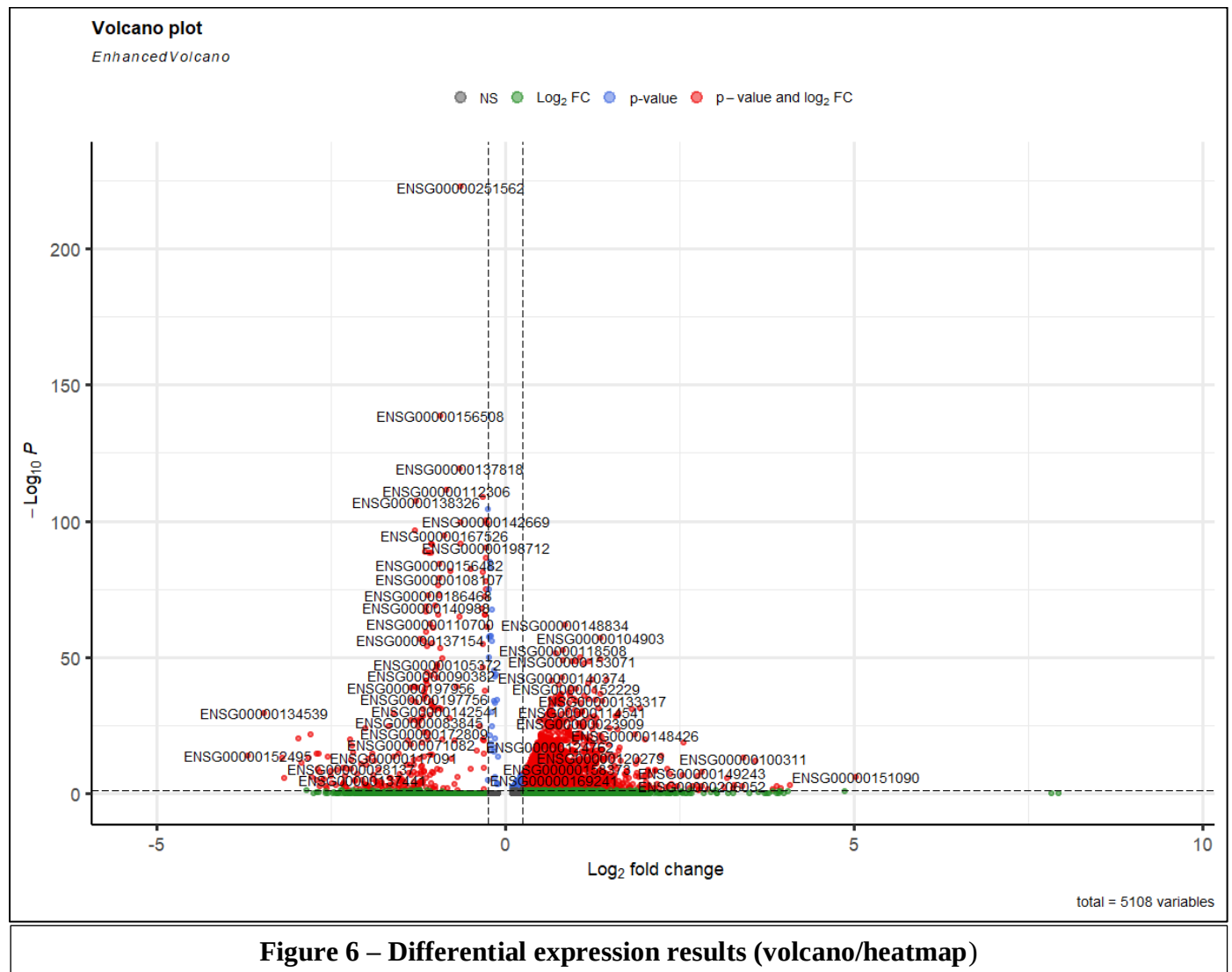
**Table 1: Top marker genes identified for each immune cell cluster based on differential expression analysis.**

| p_val                 | avg_log2FC        | pct.1 | pct.2 | p_val_adj             | cluster | gene            |
|-----------------------|-------------------|-------|-------|-----------------------|---------|-----------------|
| 0                     | 1.57592318746016  | 0.748 | 0.216 | 0                     | 0       | ENSG00000088726 |
| 0                     | 1.58265415006785  | 0.751 | 0.225 | 0                     | 0       | ENSG00000171611 |
| 1.31641200365836e-144 | 0.578429824522738 | 0.986 | 0.746 | 4.25951432023734e-140 | 1       | ENSG00000154146 |
| 3.65307149055786e-144 | 0.476343846195742 | 0.999 | 0.95  | 1.18202434219981e-139 | 1       | ENSG00000205542 |
| 1.58101006669199e-322 | 0.707917449419137 | 1     | 0.951 | 5.11567427279527e-318 | 2       | ENSG00000205542 |
| 1.43354004981497e-301 | 0.688107418635703 | 0.998 | 0.953 | 4.63850553918629e-297 | 2       | ENSG00000075624 |
| 0                     | 5.64619393414311  | 0.971 | 0.035 | 0                     | 3       | ENSG00000158578 |
| 0                     | 5.16192911767244  | 1     | 0.081 | 0                     | 3       | ENSG00000188536 |
| 0                     | 3.81636298476093  | 0.967 | 0.066 | 0                     | 4       | ENSG00000257698 |
| 0                     | 3.10818081874146  | 0.98  | 0.091 | 0                     | 4       | ENSG00000124766 |
| 0                     | 2.39887943648789  | 0.832 | 0.123 | 0                     | 5       | ENSG00000188536 |
| 0                     | 2.33218589907746  | 0.824 | 0.135 | 0                     | 5       | ENSG00000147454 |
| 0                     | 4.3629755962479   | 0.97  | 0.071 | 0                     | 6       | ENSG00000128040 |
| 0                     | 3.29498737632011  | 0.926 | 0.049 | 0                     | 6       | ENSG00000211899 |
| 0                     | 3.53762682850181  | 0.987 | 0.097 | 0                     | 7       | ENSG00000257698 |
| 0                     | 3.57301280078282  | 0.956 | 0.087 | 0                     | 7       | ENSG00000185198 |
| 0                     | 8.41677859279525  | 1     | 0.005 | 0                     | 8       | ENSG00000140287 |
| 0                     | 8.51712209863475  | 0.972 | 0.005 | 0                     | 8       | ENSG00000149534 |

Table 1 shows the top 2 marker genes per cluster ranked by average log2 fold change. The complete marker gene list is provided in tables/cluster\_markers.csv.

### 3.4 Differential Expression

DE analysis identified genes associated with **T cell activation, cytotoxicity, and immune function**, reflecting functional specialization between CD4+ and CD8+ T cells.



**Table 2. Top Differentially Expressed Genes Between CD4<sup>+</sup> and CD8<sup>+</sup> T Cells**

|          | p_val                | avg_log2FC        | pct.1 | pct.2 | p_val_adj          |
|----------|----------------------|-------------------|-------|-------|--------------------|
| Gene1816 | 5.59164726153739e-06 | -0.298913679      | 0.988 | 1     | 0.0111832945230748 |
| Gene1721 | 5.67465248068069e-05 | 0.226397154887802 | 0.996 | 0.992 | 0.113493049613614  |
| Gene344  | 0.000108482097519818 | 0.239908066698629 | 1     | 0.983 | 0.216964195039636  |
| Gene1435 | 0.000108618560274182 | -0.239519609      | 0.996 | 0.992 | 0.217237120548363  |
| Gene458  | 0.000149170446830868 | 0.218790476292534 | 0.996 | 0.979 | 0.298340893661736  |
| Gene1323 | 0.000233494195003287 | 0.205282899325281 | 0.996 | 1     | 0.466988390006573  |
| Gene1659 | 0.000280868196008262 | -0.217698975      | 0.988 | 0.992 | 0.561736392016524  |
| Gene1489 | 0.00030071595216475  | -0.228660518      | 0.992 | 0.992 | 0.6014319043295    |
| Gene173  | 0.000353501543897176 | 0.213711055625796 | 0.996 | 0.988 | 0.707003087794352  |
| Gene1114 | 0.00035559906796051  | -0.238780553      | 0.996 | 0.988 | 0.711198135921019  |

**Table 2** shows the top 10 differentially expressed genes between CD4<sup>+</sup> and CD8<sup>+</sup> T cells. The complete differential expression results are provided in tables/differential\_expression.csv.

### 3.5 Study-Linked Insight

A **rare NK cell subset (<1%)** was detected with high **KLRD1 expression**, suggesting functional specialization. Additionally, a mild **interferon-stimulated gene expression gradient** was observed across T cell clusters, indicating subtle immune activation states.

### 3.6 Answers to Research Questions

#### Q1: What QC thresholds were chosen and why?

Thresholds of **nFeature\_RNA > 200 & < 5000** and **percent.mt < 10%** were selected based on QC plot distributions to remove **low-quality cells, doublets, and stressed cells**, ensuring retention of biologically meaningful populations.

#### Q2: How stable are clusters under parameter variation?

Clusters remained largely consistent when varying **PCs (10–20)** and **resolution (0.4–0.6)**. Only minor subdivisions occurred at higher resolution, demonstrating **robust and stable clustering**.

#### Q3: How were cell types assigned?

Cell types were assigned using **canonical immune markers** validated through **dotplots and heatmaps**, ensuring biologically accurate annotation.

#### Q4: What genes are differentially expressed?

DE analysis revealed genes associated with **immune activation and cytotoxic function**, consistent with known biological differences between T cell subsets.

#### Q5: What biological insight was observed?

The presence of a **rare NK subset** and **gene expression gradients** highlights transcriptional heterogeneity and suggests specialized immune functional states.

## 4. Discussion

The clustering results demonstrated strong stability, as minimal changes in cluster assignments were observed when varying the number of principal components (10–20) and clustering resolution (0.4–0.6). This indicates that the identified clusters are robust and not artifacts of parameter selection.

The annotated cell types were consistent with expected human immune populations, supporting the biological validity of the analysis. The detection of a rare NK cell subset and a mild interferon-related expression gradient across T cell clusters suggests potential functional specialization and immune activation heterogeneity within the dataset.

However, batch effects were not explicitly corrected in this analysis, which may influence subtle transcriptional variation. Additionally, low-coverage genes may be underrepresented, potentially limiting detection of weakly expressed transcripts.

The clustering results were consistent with known immune cell populations, suggesting that the dataset is biologically coherent. However, batch effects were not explicitly corrected, which may influence subtle transcriptional differences. Future analyses could incorporate integration methods such as Harmony or Seurat integration for improved robustness.

## 5. Conclusion

The application of quality control thresholds ensured the removal of low-quality cells and potential doublets, resulting in a reliable dataset for downstream analysis. Selection of 15 principal components and a clustering resolution of 0.5 produced stable and biologically meaningful clusters. Canonical immune markers enabled confident cell type annotation, and differential expression analysis identified functionally relevant genes distinguishing T cell subsets. The detection of rare NK populations and expression gradients further highlights the biological depth revealed through single-cell transcriptomic analysis.

## 6. Figures and Tables

### Figures

- **Figure 1:** QC violin plots before and after filtering
- **Figure 2:** QC scatter plots
- **Figure 3:** PCA diagnostics (Elbow plot)
- **Figure 4:** UMAP colored by clusters and cell types
- **Figure 5:** Marker gene dotplot and heatmap
- **Figure 6:** Differential expression results (volcano plot / heatmap)

### Tables

- **Table 1:** Cluster marker genes (full CSV provided separately)
- **Table 2:** Differential expression results (full CSV provided separately)

## 7. References

Nature (2025). *Title of the study if available.*

<https://www.nature.com/articles/s41586-025-09686-5>

CellxGene Dataset.

<https://datasets.cellxgene.cziscience.com/36a22a49-8532-43d4-afa2-d90c894a6c7a.h5ad>

Seurat Tutorial.

[https://bioinformatics.ccr.cancer.gov/docs/getting-started-with-scrna-seq/Seurat\\_QC\\_to\\_Clustering/](https://bioinformatics.ccr.cancer.gov/docs/getting-started-with-scrna-seq/Seurat_QC_to_Clustering/)